

K.J. Somaiya College of Engineering, Mumbai-77

(Constituent college of Somaiya Vidyavihar University)

DEPARTMENT OF MECHANICAL ENGINEERING

Mini Project Report

on

**Effect of Accessories on Aerodynamics
of Vehicle**

by

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Batch 2020 -2024

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DEPARTMENT OF MECHANICAL ENGINEERING

Certificate

This is to certify that the Mini project report entitled '**Effect of Accessories on aerodynamics of vehicle**' is bonafide record of the work done by **Sahil Pujari (16010520045), Pravin Salla (16010520046) & Ayush Shah (16010520052)** in the year **2022-2023** under the guidance of **Prof. Prashant Jain and Prof. Vijay Shinde**.

Guide / Co-Guide

Head of the Department

Principal

Date:

Place: Mumbai-77

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Date:

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Abstract

In this era of rapidly depleting natural resources, the rise in fuel prices continues to rise. Car manufacturers have strict standards regarding environmental policy on the way to producing efficient vehicles with lower emissions. High speed cars are at stake to provide uncompromising performance, but they have strict rules emissions force companies to take a different approach to sustain performance requirements intact.

One of the most sought-after paths is improvement vehicle aerodynamics. Drag force is one of the main obstacles when the vehicle is in motion. This Project aims to investigate the effects of various accessories/add-ons on a vehicle to reduce drag and to make the vehicle more aerodynamic and overall, a more efficient vehicle and as a result, the fuel consumption is also improved.

The three-dimensional car model is developed in SOLIDWORKS v22. Computational Fluid Dynamics (CFD) is performed to understand the effects of these additional devices. CFD is performed in the ANSYS Fluent module. Drag coefficient (C_D), Lift Coefficient (C_L), Drag Force and Lift Force are calculated and compared in several cases. The result of the simulations was analyzed and observed that different devices represented several different functions but maximum resistance reduction was found in the case of GT with spoiler and diffuser.

Key words: Aerodynamics, CFD, Drag coefficient, Drag force

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Nomenclature

$V \rightarrow$ Velocity of the body

$F_D \rightarrow$ Drag force

$F_L \rightarrow$ Lift force

$\rho \rightarrow$ Density of the fluid medium (air)

$A \rightarrow$ Frontal area of the body facing the fluid

$C_D \rightarrow$ Coefficient of drag of the body

$C_L \rightarrow$ Coefficient of lift of the body

Chapter 1

Introduction

This chapter presents an overview of car aerodynamics, the forces and factors that are involved in it. Also why is it necessary to analyze the aerodynamics of a vehicle.

1.1 History

It was understood intuitively, if not scientifically since the dawn of the automobile that air presented the greatest obstacle to automotive speed and economy putting it into practice was quite another story. The frictional force of aerodynamic drag increases significantly with vehicle speed. As early in the 1920s engineers had to consider automobile shape in reducing drag at higher speeds.

By the 1950s German automotive engineers were thoroughly analyzing the effects of automotive drag for the higher performance of vehicles. By late 1960s engineers also became aware of the significant increase in sound levels emitted by automobiles at higher speeds. These effects were understood to increase the intensity of sound levels for adjacent land used at a non-linear rate.^[2]

Highway engineers started designing roadways considering the effects of speed and aerodynamic forces produced sound levels, and automobile manufacturers also considered similar factors during vehicle design.

1.2 Theory

The study of how gases interact with moving bodies is known as aerodynamics. Aerodynamics is mainly concerned with the forces drag and lift that are induced by air flowing through and over solid bodies since air is the most common gas.^[2] Engineers use aerodynamic concepts during the design of a wide range of objects, including bridges, houses and even golf balls; however, the aerodynamics of a aeroplane and automobiles are of their primary concern.

Aerodynamics is also used in the study of aeronautics, which is the science of constructing and flying an aircraft. Engineers design aircrafts that can navigate through the Earth's atmosphere using aerodynamic principles. Weight, lift, thrust, and drag are the four

principles of aerodynamics. These physics of flight and aircraft structures forces can be compared to the forces that act on a car on the road.^[1]

Weight: There is a weight to everything on our planet. Gravity pulls things downwards. A plane must be propelled in the opposite direction of weight in order to lift. The force needed to push an object is ruled by its weight. In comparison to a plane, a kite needs much less upward thrust.

Lift: A lift is a force that causes something to move upwards. It is the exact opposite of gravity. An aircraft's lift has to be greater than its weight in order for it to rise. The wings of an aeroplane provide lift. It is the shape of an aeroplane's wings that allows it to fly. The top of a plane's wing is curved, while the bottom is flatter. Because of this form, air flows faster over the top and slower under the bottom. As a result, the air pressure is less on top of the wing. The wing, as well as the aeroplane to which it is connected, moves up. The sail of a boat is similar to a wing. That is what propels a sailboat.

Drag: A force that seeks to slow something down is called drag. It makes moving an object difficult. Walking or running through water is more difficult than walking or running through on land. This is due to the water having a greater drag coefficient than air. The amount of drag produced is affected by the shape of an object. When compared to flat surfaces, most spherical surfaces will have less drag. Surfaces that are narrow will have less drag than those that are broader. The more air that comes into contact with any surface, the more drag it will produce.

Thrust: The power of thrust is the exact opposite of drag. Thrust is the forward movement of something. An aircraft must have more thrust than its drag to keep moving forward. A propeller can provide thrust to a small jet. Jet engines could provide propulsion to larger vehicles. There is no thrust in a glider. It can only fly until the drag slows it down and forces it to land.

These above factors are taken in to consideration when designing any object whose aerodynamics has an effect on its function. As we saw the role of aerodynamics is a major one, and that makes it important to test the aerodynamics of the parts or object you design.

The advantages of an aerodynamically well designed vehicle are:-

Speed – The frictional force of aerodynamic drag increases significantly with speed of the vehicle. Therefore, a vehicle is designed to minimize drag and capable of achieving greater speeds with less effort.

Safety – An aerodynamically sound vehicle body design ensures greater aerodynamic stability and achieving greater straight-line and cornering speeds safely and confidently is made easier.

Fuel Efficiency – Streamlined vehicles require less power to overcome air resistance and are, therefore, more fuel efficient.

Fuel Economy – Better fuel efficiency ensures more economical vehicle to operate.

Damage – Due to less drag there would be less wear and tear on the body, thus reducing repair and damage costs.

1.3 Objective

- To create a three-dimensional Computer Aided Design (CAD) model of the outer body shape of the first-generation Koenigsegg Jesko Absolut using Dassault Systèmes Computer Aided Three-Dimensional Interactive Application (SOLIDWORKS v23) software and blueprints.
- To measure the aerodynamic performance parameters of lift and drag of the standard outer body shape of the Koenigsegg Jesko Absolut by simulating suitable aerodynamic conditions around its CAD model in commercial Computational Fluid Dynamics (CFD) software ANSYS FLUENT.
- To find the aerodynamic performance parameter measurements under identical aerodynamic conditions in ANSYS FLUENT® after attaching aerodynamic accessories individually and in combination as well to the CAD model of the Koenigsegg Jesko Absolut in SOLIDWORKS v23.

Case 1 – Baseline GT Model

Case 2 – With Lip Spoiler at rear

Case 3 – With Rear Wing

Case 4 – With Diffuser

Case 5 – With Fins

Case 6 – With Lip Spoiler and Diffuser

Case 7 – With Rear Wing and Diffuser

Case 8 – With Fins and Diffuser

- To study the pressure contours and velocity streamlines in the flow fields generated in each of the eight cases in order to determine the causes for the variation in the values of the aerodynamic performance parameters of lift and drag measured and to, therefore, compare the results of the simulation tests run on all eight cases and determine the most aerodynamically efficient configuration of the eight.

Chapter 2

Literature Survey

This chapter presents the basics of aerodynamics and the effects of several accessories on aerodynamics. It also goes in depth on some important concepts such as flow separation and wake etc.

Drag Force generally depends upon the frontal area, velocity and drag coefficient of the body. It can be expressed as:

$$F_D = 0.5 C_D \rho A V^2$$

Where F_D is the drag force, C_D is the drag coefficient of the body, ρ is the density of fluid medium, A is area of body facing the fluid, V is velocity of the body.^[3]

In a similar sense, the lift force is also a major concern for designers, as excess lift can cause the vehicle to lose traction at high speeds resulting in injuries for both the driver and onlookers along with damage of property. Therefore, it is highly desirable that the lift should be within the calculated region.

Lift force can be expressed as:

$$F_L = 0.5 C_L \rho A V^2$$

Where F_L is the lift force, C_L is the coefficient of lift of the body, A is area of the body facing the fluid; V is the velocity of the body, ρ is the density of fluid medium which is air.^[3]

From the above equations, it can be seen that the drag force is proportional to the square of velocity. This implies that the resistance caused by air increased exponentially with the speed of the body. Flow separation control is also of certain concern in fundamental fluid dynamics and other engineering applications.

Flow Separation determines the size of the wake area and the aerodynamic drag is resolved accordingly. When the air moving around the vehicle is separated at the rear end, it leaves a large low-pressure turbulence behind known as the wake. This wake is the reason of

formation of pressure drag. Several techniques have been experimented to control the flow separation either by reducing its effects or preventing it.^[4]

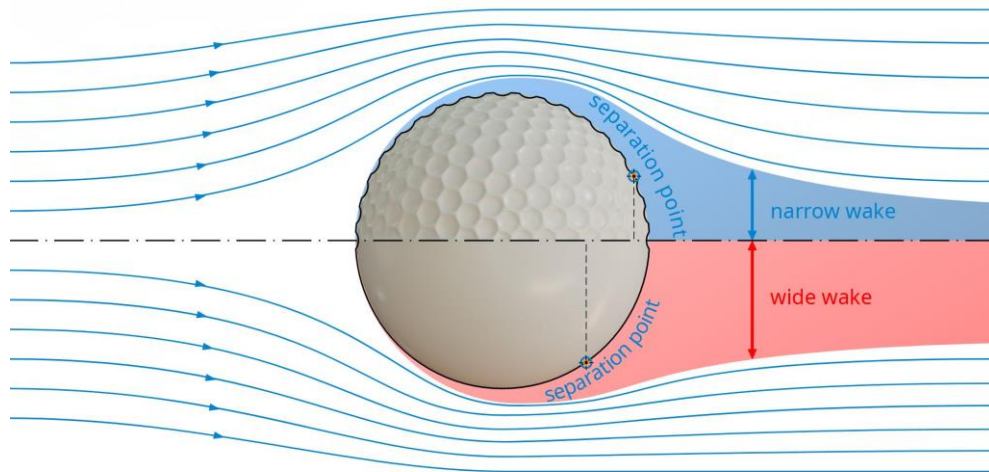


Figure 2. 1 Flow separation and formation of wake region

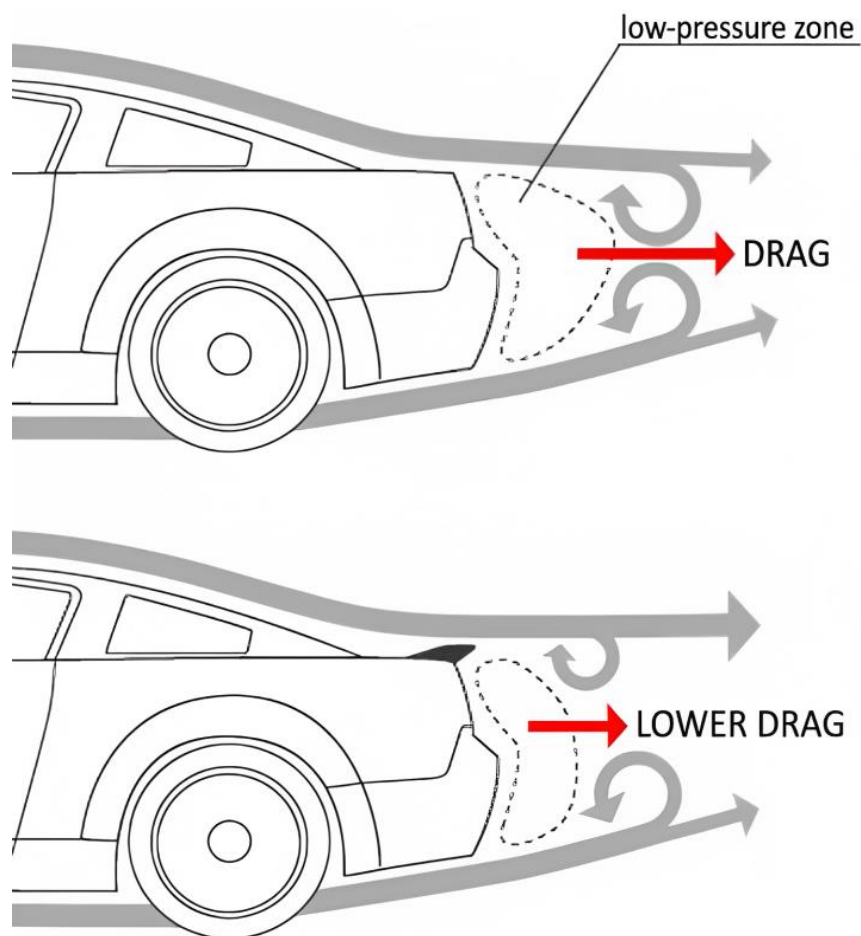


Figure 2. 2 Effect on drag by using spoiler

Spoiler is the most widely used aerodynamic device in the automotive industry. Its main motive is to spoil the unwanted air and channel the airflow in order, which will help in reducing drag. Production vehicles adopt this method to increase the appearance appeal of the vehicle, which provides little to none of the aerodynamic advantages. Thus, mostly high speed vehicles adapt it to achieve greater speeds.^[1] The low-pressure zone behind the vehicle is reduced, thus turbulence is decreased, which subsequently leads to drag reduction.



Figure 2. 3 Car with spoiler

The wing is an indispensable aerodynamic device often used on racing vehicles. A rear wing might look like a larger spoiler but is different in its functions. It is shaped like the wing of a plane but upside down. Its main motive is to provide ample downforce or negative lift so the vehicle will have increased traction and the vehicle stays grounded.



Figure 2. 4 Car with wing

It allows the vehicle to corner better and improves its stability at high speeds. But using a wing will increase the drag to the vehicle. Thus, for lift gained, drag will also increase. It is generally considered as a tradeoff between lift or drag.^[3]

For the first time in the automotive space, the use of fins at the rear part of the vehicle body is seen by Swedish hypercar manufacturer Koenigsegg Automotive AB. Their latest model the “Jesko Absolut” having the least drag coefficient in their lineup has fins used at its rear as shown in Fig 2.5. Fins are inspired by fighting jets that provide high-speed stability as well as reduce the aerodynamic drag.



Figure 2. 5 Koenigsegg Jesko Absolut

The diffuser is one of many aerodynamic devices found on F1 cars. The wide versatility offered by diffusers has helped it in its way down to high speed street legal vehicles.

Diffusers are competent of reducing drag as well as increasing downforce for vehicles. The function of the diffuser is to expand the airflow from below the car through to the rear, and this will produce a pressure potential that causes ground effect, which accelerates the airflow below the car, developing vacuum like conditions below the car body.



Figure 2. 6 Car with Diffuser

The working principle behind the diffusers is derived from Bernoulli's principle which states that, A slow moving fluid exerts a greater pressure than a fast moving fluid. Therefore, the motive of diffusers is to accelerate the airflow underneath the car so that less pressure is maintained in comparison to the airflow around the outer body. This helps in shooting out the air from underneath the car. The diffuser then helps in reducing this high speed air down to ambient speed and help in filling the area at the rear of the vehicle, making the underbody an important downforce generating and also importantly reducing the drag of the vehicle.^[2]

Chapter 3

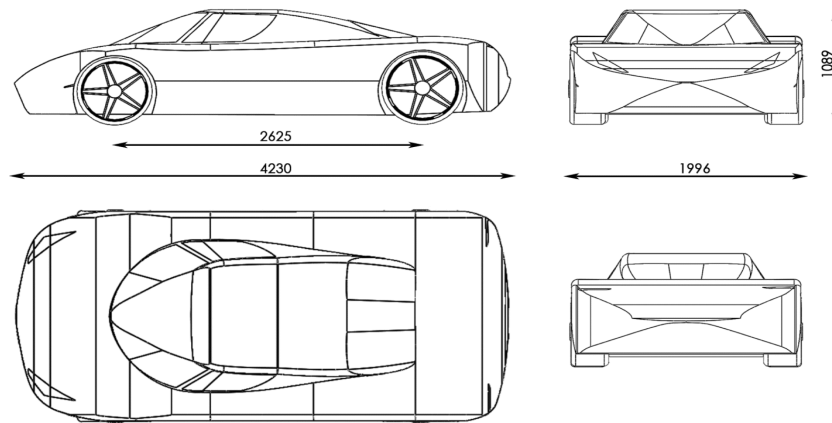
Methodology

This chapter presents the entire process of doing CFD on a car. Right from the beginning from designing a 3D model to simulation and all considerations that need to be taken into account during the process.

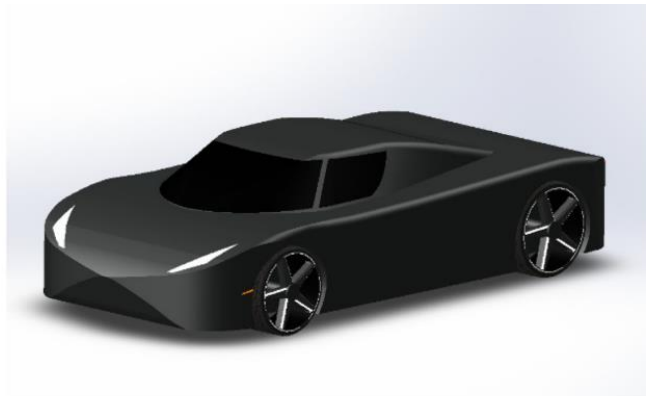
3.1 Geometric modelling

3.1.1 Baseline GT model

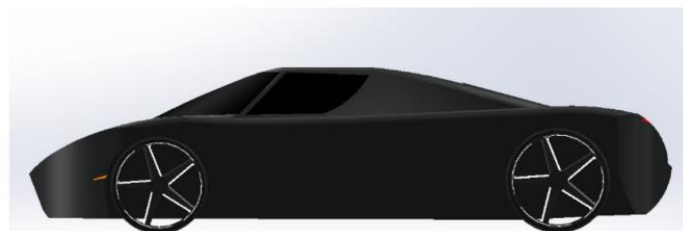
The three-dimensional car model was developed in SOLIDWORKS v23.0. The stock GT model's length, width, height is 4320 mm, 1990 mm, 1090 mm respectively. While the ground clearance is 90 mm.



(a) Dimensions Blueprint



(b) Isometric view

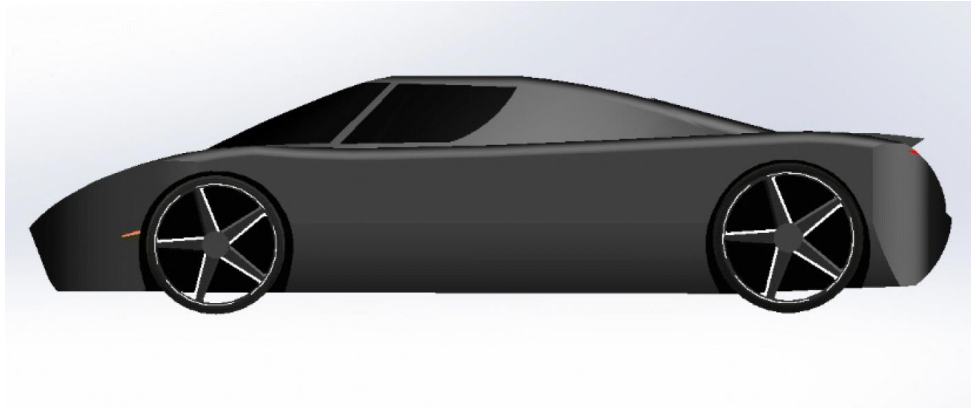


(c) Side view

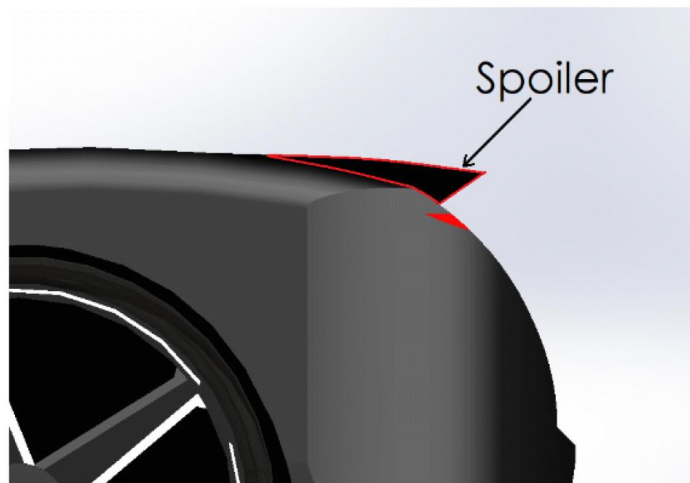
Figure 3. 1 Geometric Model of Baseline GT

3.1.2 GT with spoiler

In the stock GT model, a spoiler is installed to the rear end of the car to generate greater streamlined flow by delaying the flow separation to reduce the overall coefficient of drag of the car.



(a) Side view



(b) Spoiler

Figure 3. 2 GT with spoiler

3.1.3 GT with wing

A wing with 1700 mm width span has been attached to the stock model at the rear of the vehicle as shown in with an attack angle of 33.9 degrees. It is expected that the rear wing will increase downforce in trade off to the drag force.

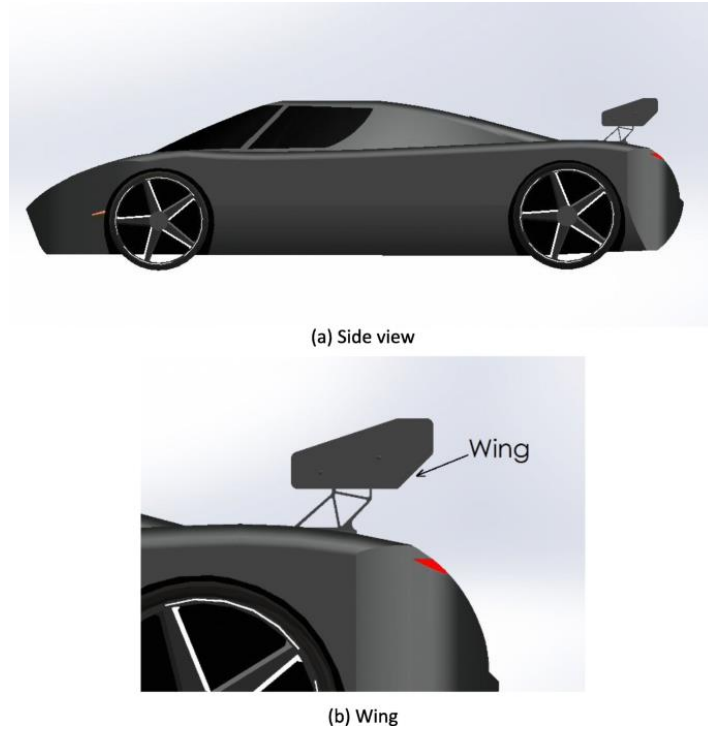


Figure 3. 3 GT with wing

3.1.4 GT with diffuser

A diffuser is developed of length 1000 mm while its angle of inclination is 17 degrees, the width of each blade is 7.5 mm. A steeper inclination angle might result in flow separation, that would lead to an increase in drag. A lower inclination angle would be less effective for our desired motive. Diffuser is attached at the rear of the stock GT.

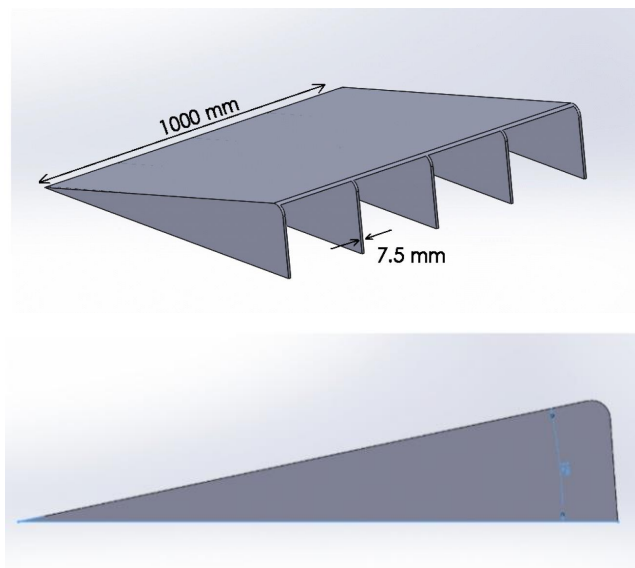


Figure 3. 4 Geometric model of diffuser

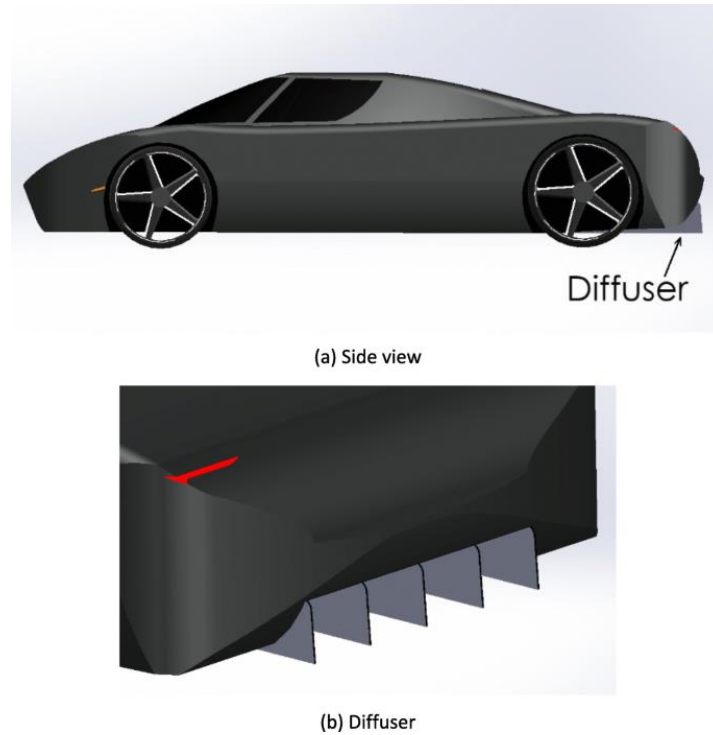


Figure 3. 5 GT with diffuser

3.1.5 GT with fins

The fins are attached to the mid-rear of the stock GT, each of thickness 25 mm to provide increase in stability at high speeds. It is expected that the overall coefficient of drag of the vehicle will decrease by the use of these fins.

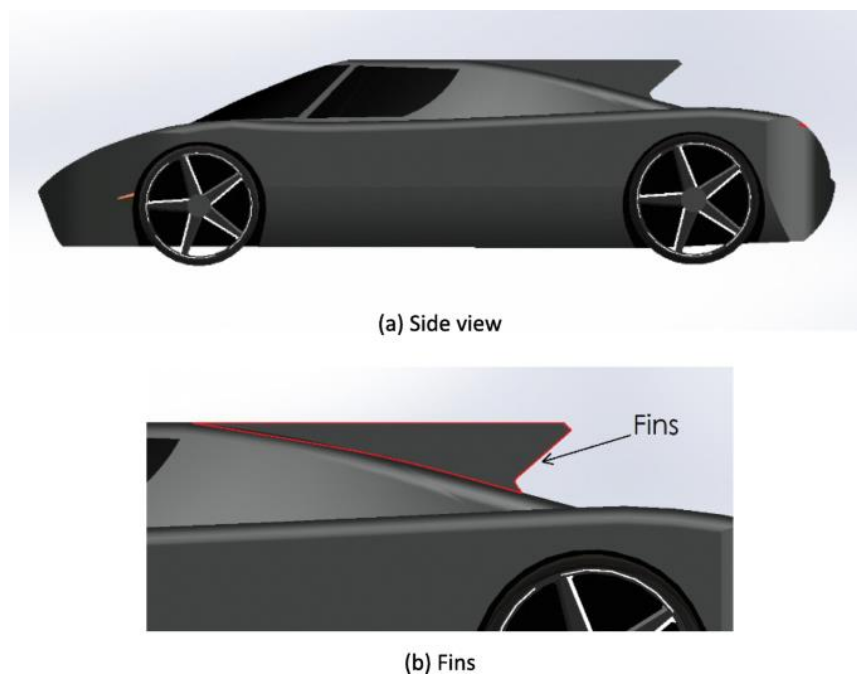


Figure 3. 6 GT with fins

3.1.6 GT with spoiler and diffuser

In the stock GT model, both diffuser and spoiler are attached. It is expected that the combined aerodynamic devices will reduce the coefficient of drag of the vehicle.

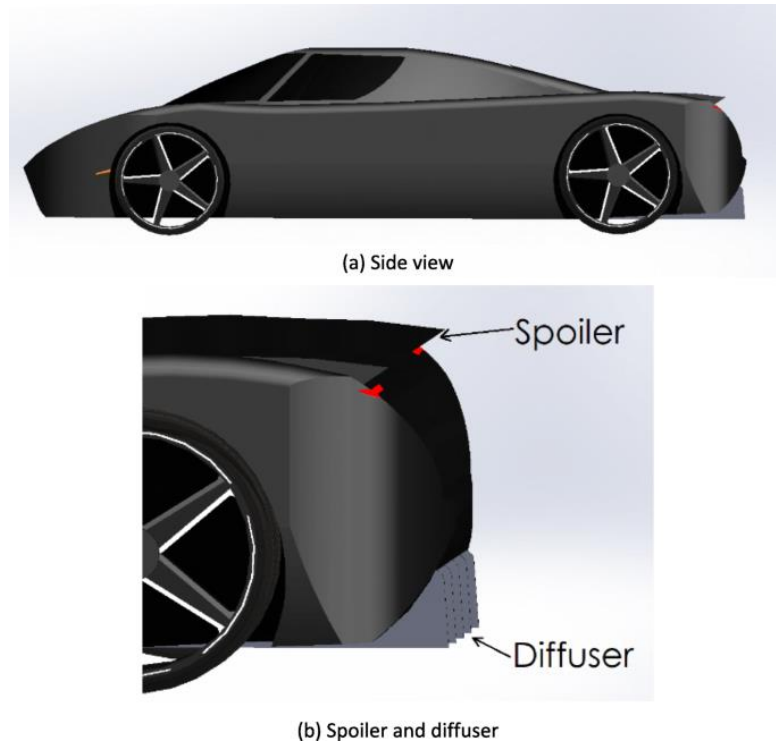


Figure 3. 7 GT with spoiler and diffuser

3.1.7 GT with wing and diffuser

In the stock GT model, both diffuser and wing are attached. Use of both of these aerodynamic devices will help in increased downforce as well as drag reduction.

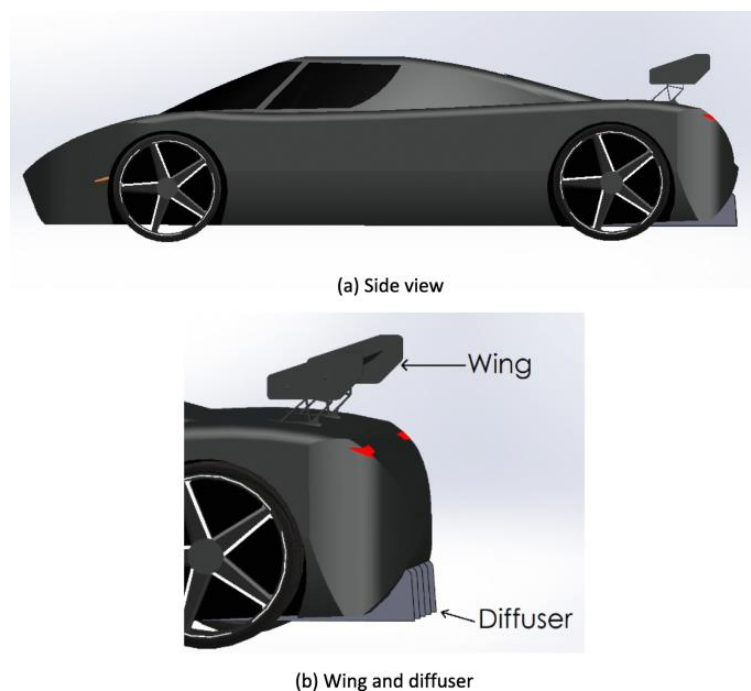


Figure 3. 8 GT with wing and diffuser

3.1.8 GT with fins and diffuser

In the stock GT model, both diffuser and fins are attached. It is expected that the use of these aerodynamic devices will help in providing stability and decreasing the coefficient of drag.

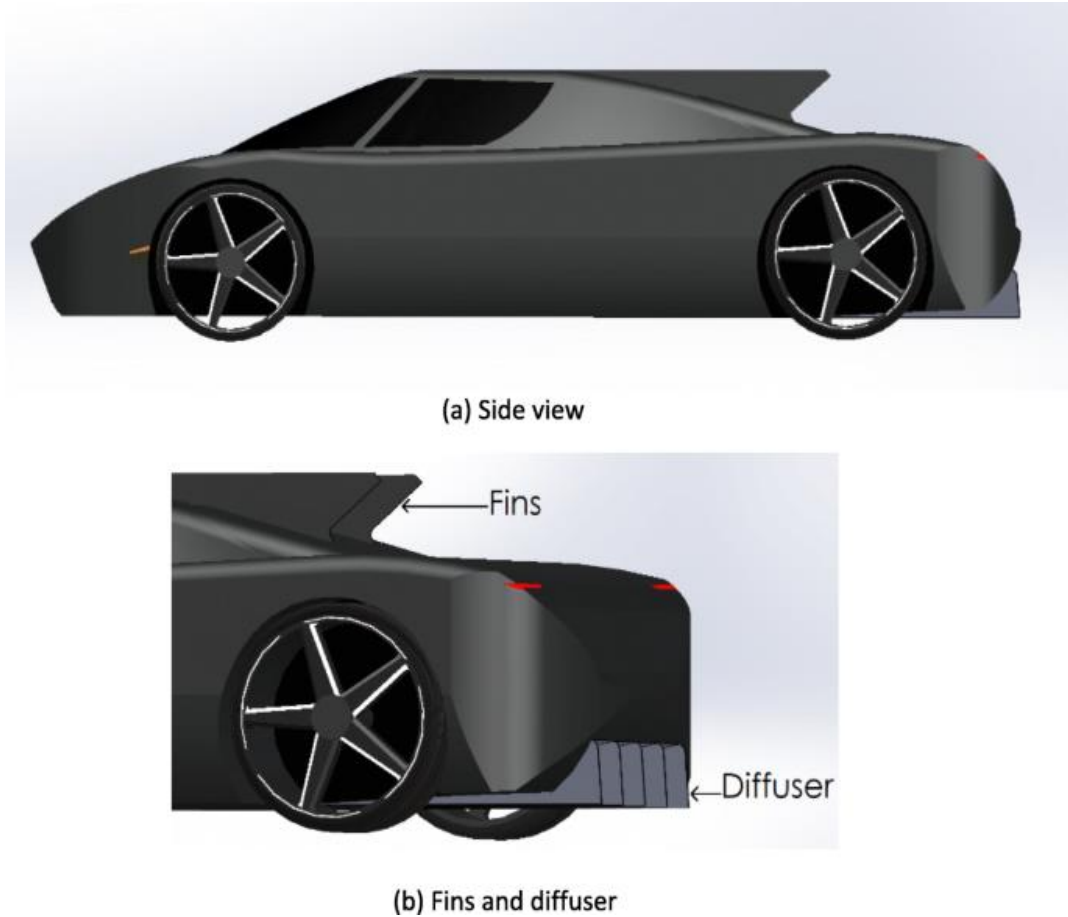
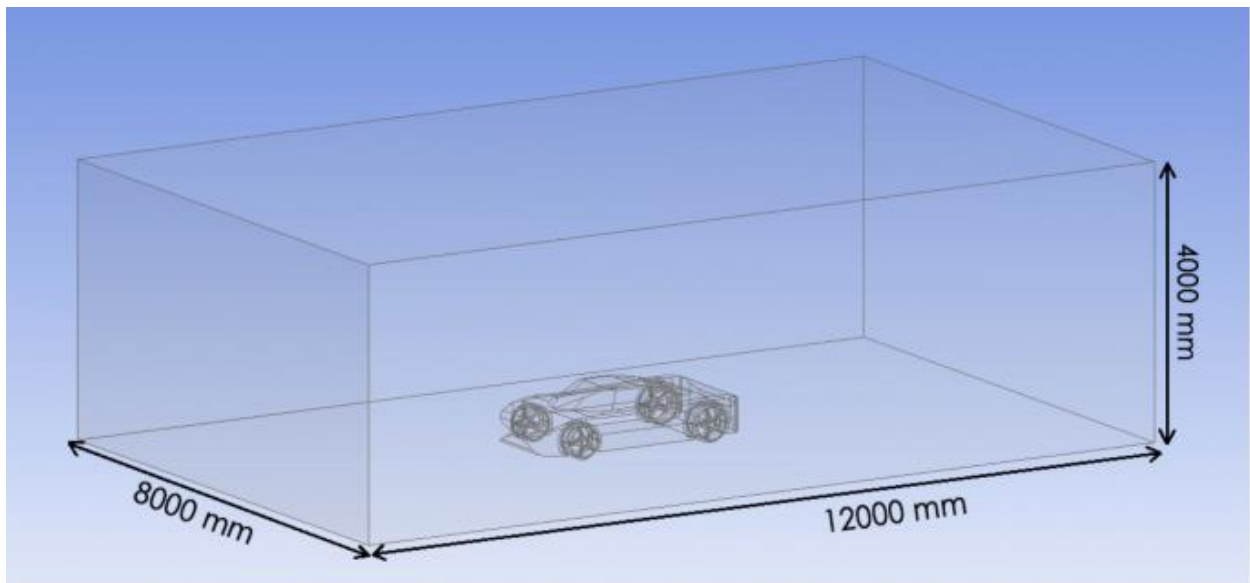


Figure 3. 9 GT with fins and diffuser

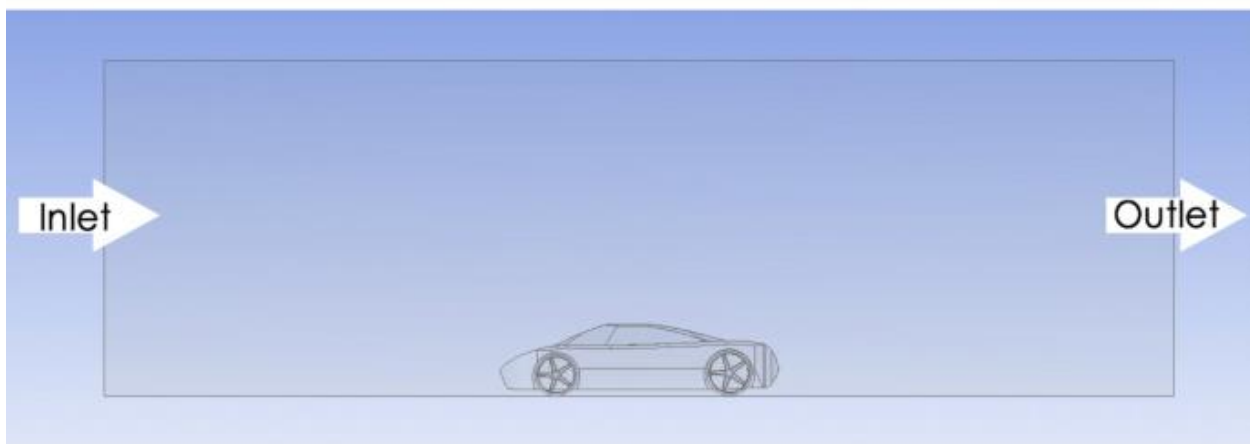
3.2 Simulation

3.2.1 Control Volume

The three dimensional car model was generated in ANSYS. Computational Fluid Dynamics (CFD) is to be carried out in the FLUENT module. In Space Claim, an enclosure is generated of dimensions 12000×4000×8000 mm to form a simulation wind tunnel where the testing will be done.



(a) Isometric View



(b) Side View

Figure 3. 10 GT in Control Volume

3.2.2 Meshing

A comprehensive mesh was generated using ANSYS Meshing Tool. The mesh was observed to be medium at the inner domain and finer near the contact region of the stock GT as shown in Fig. 3.11. A total of 2,32,186 nodes and 10,98,393 elements are formed while the meshing was generated.

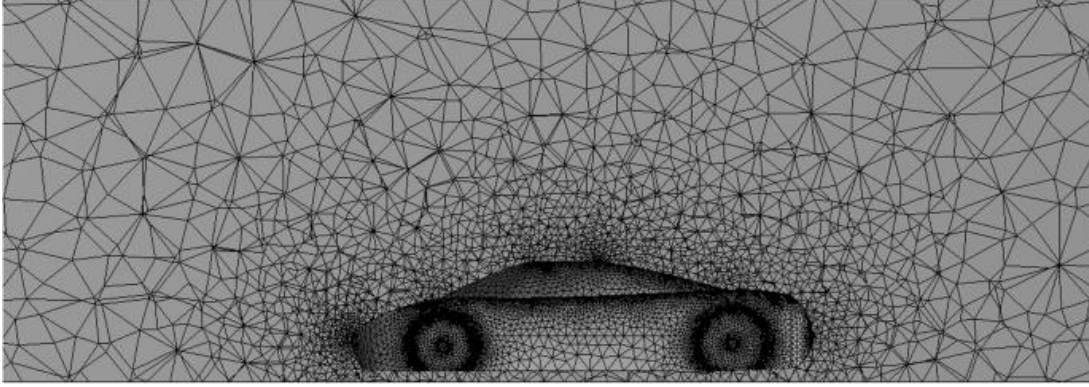


Figure 3. 11 Mesh of GT and enclosure

3.3 Boundary conditions

The numerical simulation was done in commercial code FLUENT. Due to its stability and ease of convergence, the Standard k-epsilon model was selected as a turbulence model. In most high-Reynolds-number flows, such as in this particular research, the wall function approach substantially saves computational resources, because the viscosity-affected near-wall region, in which the solution variables change most rapidly, does not need to be resolved. The wall function approach is popular because it is economical, robust, and reasonably accurate. It is a practical option for the near-wall treatments for industrial flow simulations. The turbulence kinetic energy, k , and its rate of dissipation, epsilon ε , are obtained from the following transport equations.

For turbulent kinetic energy k

$$\frac{\partial(\rho k)}{\partial t} + \frac{\partial(\rho k u_i)}{\partial x_i} = \frac{\partial}{\partial x_j} \left[\frac{\mu_t}{\sigma_k} \frac{\partial k}{\partial x_j} \right] + 2\mu_t E_{ij} E_{ij} - \rho \varepsilon$$

For dissipation ε

$$\frac{\partial(\rho \varepsilon)}{\partial t} + \frac{\partial(\rho \varepsilon u_i)}{\partial x_i} = \frac{\partial}{\partial x_j} \left[\frac{\mu_t}{\sigma_\varepsilon} \frac{\partial \varepsilon}{\partial x_j} \right] + C_{1\varepsilon} \frac{\varepsilon}{k} 2\mu_t E_{ij} E_{ij} - C_{2\varepsilon} \rho \frac{\varepsilon^2}{k}$$

In these equations, u_i represents the velocity component in corresponding direction, E_{ij} represents the component of rate of deformation, and μ_t represents the eddy viscosity. $C_{1\varepsilon}$

and $C_{2\varepsilon}$ are the constants. ∂k and $\partial \varepsilon$ are the turbulent Prandtl numbers for k and ε respectively ^[4]. Values of constant $C_{1\varepsilon}$ is 1.33, $C_{2\varepsilon}$ is 1.82, ∂k is 1.05 and $\partial \varepsilon$ is 1.20. The coupled scheme was set as the iterative algorithm and the residual value was set to 0.0001. The frontal surface area of the vehicle is 1.9820 m². A constant velocity boundary condition was selected for the inlet boundary. Since our accessories make a difference at higher speeds, thus velocity is kept at 150 kmph. Ground was selected as moving wall to imitate the road. The outlet was set to ambient pressure condition. Boundary conditions used for the simulation are listed below in Table 3.a.

Table 3.a Boundary conditions

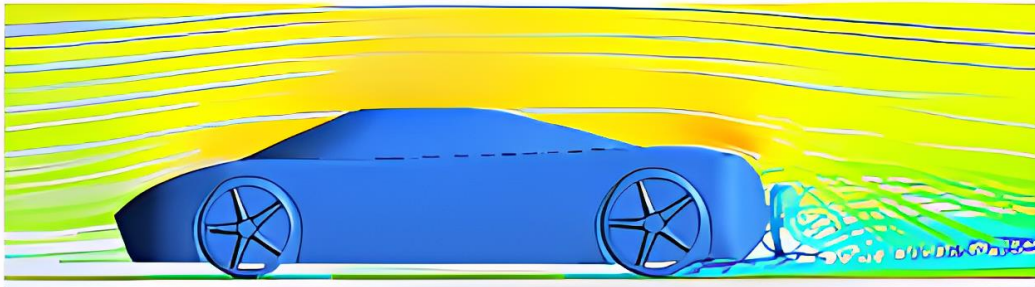
Region	Boundary condition
Inlet	Velocity inlet, $v = 150$ km/h
Outlet	Pressure outlet, reference pressure = 0 Pa
Vehicle surface	No slip wall
Top and side	No slip wall
Ground	Moving Wall, $v = 150$ km/h
Temperature	297.15 K

Chapter 4

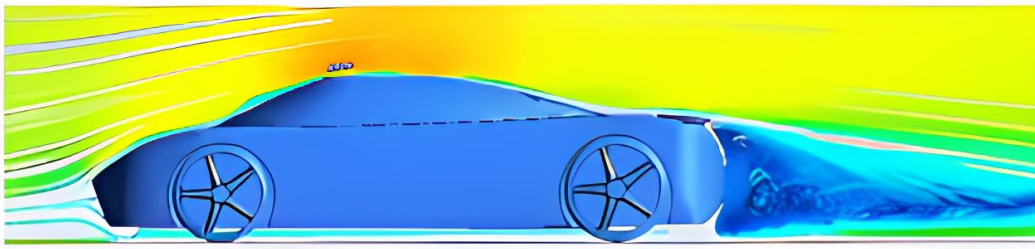
Observations

This chapter presents the different effects of accessories on wind flow around the vehicle with the help of simulation. All the data is also tabulated for a better understanding.

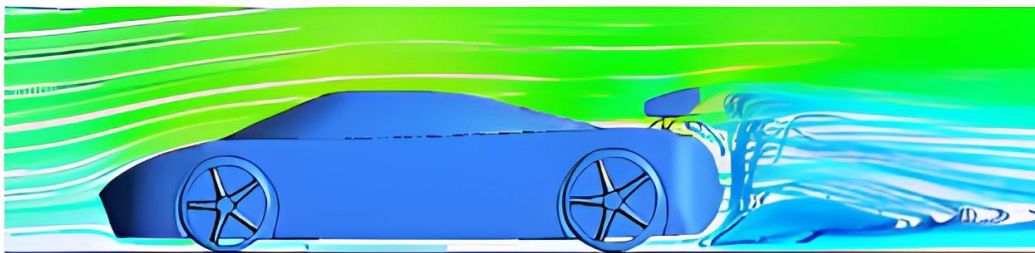
The results of the drag coefficients obtained are discussed below.



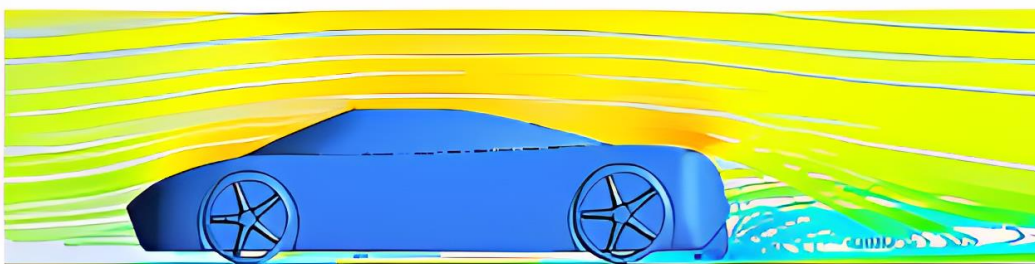
(a) GT baseline



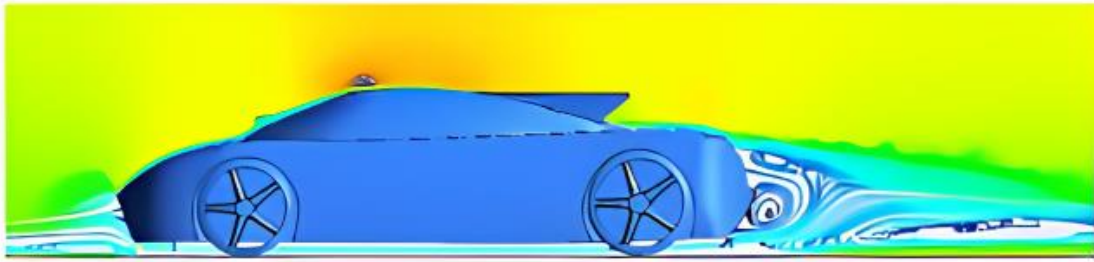
(b) GT with spoiler



(c) GT with wing



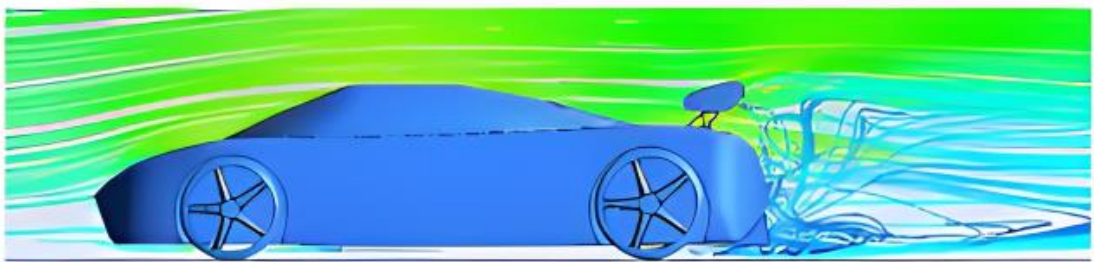
(d) GT with diffuser



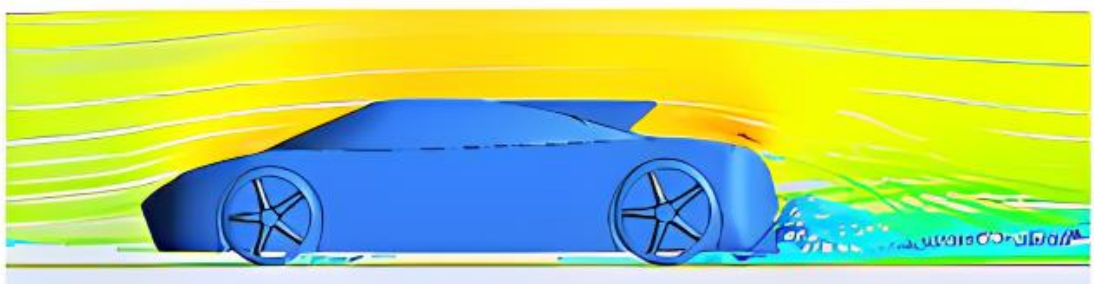
(e) GT with fins



(f) GT with spoiler and diffuser



(g) GT with wing and diffuser

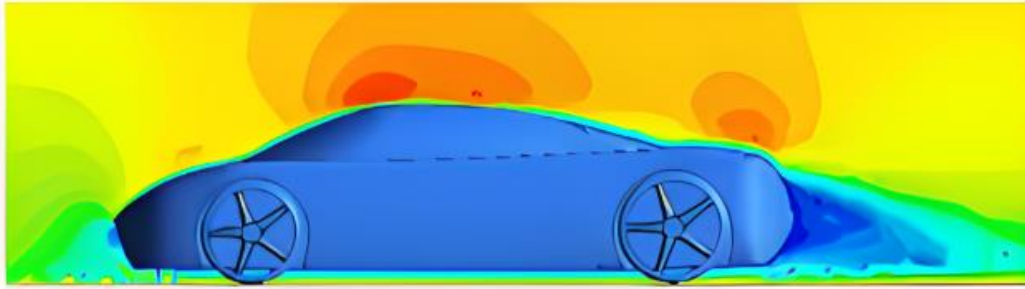


(h) GT with fins and diffuser

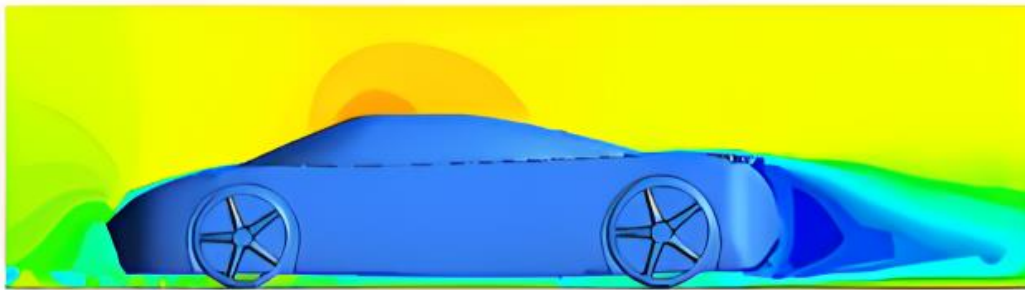
Figure 4. 1 Streamlines after vehicle

Fig.4.1 depicts the streamlines plot derived from the simulation for different cases. As seen from the streamlines plot in the figure, minimum flow separation is favourable, which subsequently leads to lesser turbulence. The amount of turbulence created behind the rear region of the car determines the magnitude of the drag force. In the case of GT with Wing Fig.4.1(c), maximum flow separation is observed, which has led to maximum turbulence,

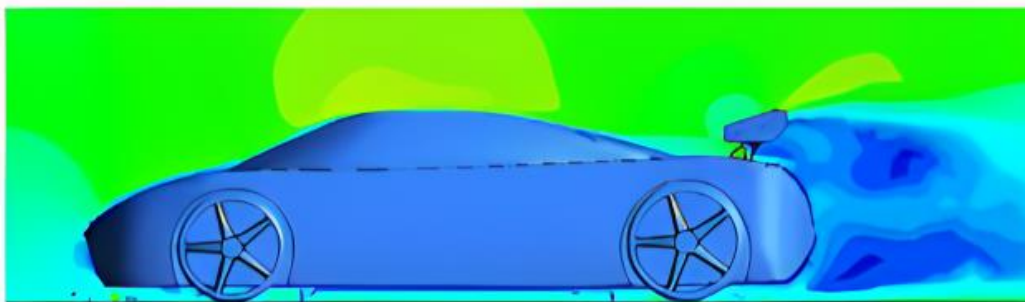
thus maximum drag force. On the other hand, the application of the Spoiler Fig.4.1(b) on the baseline model reduced the turbulence at the back, thus drag force is also reduced considerably. Moreover, the addition of Diffuser Fig. 4.1(d) to the baseline model accounted for streamlined flow, which reduced the drag force. In the combination of these two in the case of GT with Spoiler and Diffuser Fig.4.1(f), it is evident from the streamlines that the least turbulence is generated, which further reduced the drag coefficient minutely, thus the least drag force out of all the cases stated above.



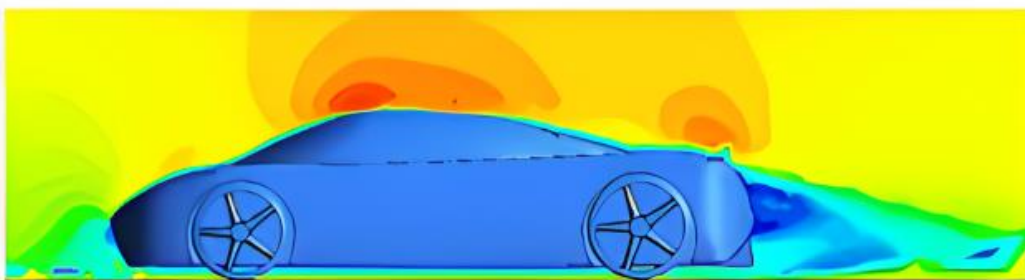
(a) GT baseline



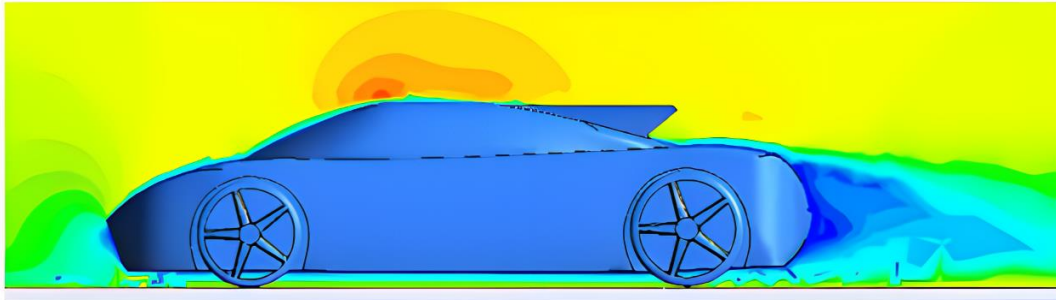
(b) GT with spoiler



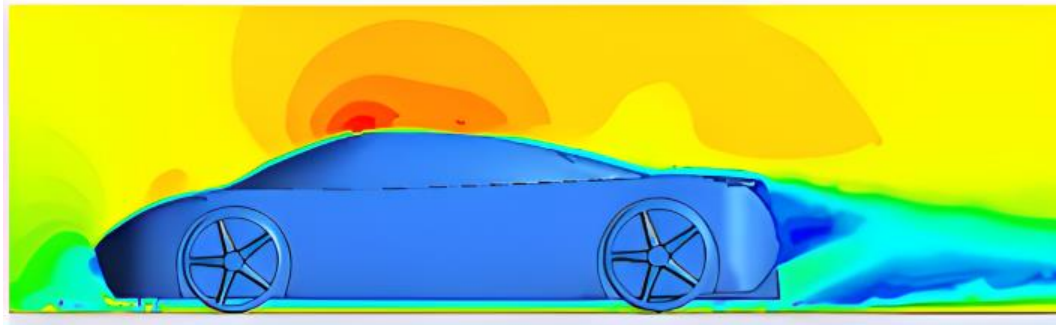
(c) GT with wing



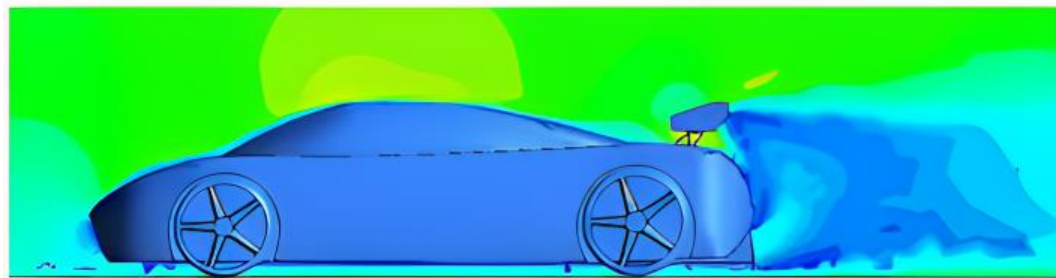
(d) GT with diffuser



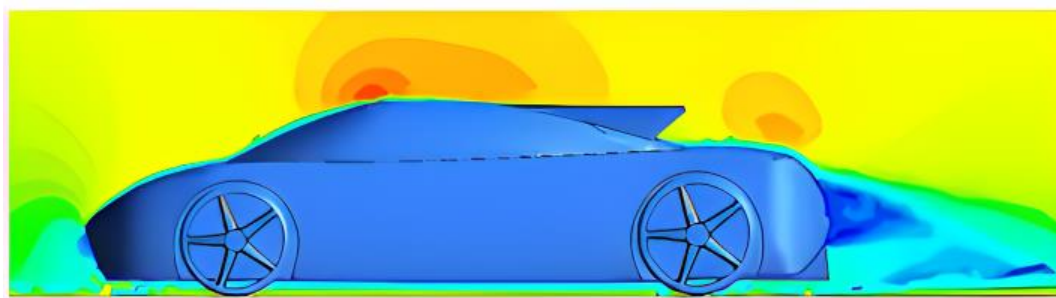
(e) GT with fins



(f) GT with spoiler and diffuser



(g) GT with wing and diffuser



(h) GT with fins and diffuser

Figure 4. 2 Velocity contours behind the vehicle

Figure 4.2 shows the velocity contour for all the cases. From velocity contours, the re-circulation zone is visualized behind the vehicle. The minimum the recirculation zone is, the least turbulence is created, which subsequently leads to minimized drag. From velocity contours derived from the simulation, it is evident that in the case of GT with Wing Fig. 4.2(c), there is a large re-circulation zone extending from the bottom of the trunk to the wing's edge. Thus, it has contributed to the maximum drag force.

Moreover, the addition of a Spoiler Fig.4.2(b), the recirculation zone is reduced from the baseline model, which reduced the drag force considerably. On the application of Diffuser Fig.4.2(d), the recirculation zone is reduced, accounting for more teardrop shape with less flow separation, which reduced the drag force. Furthermore, the combination of these two in the case of GT with Spoiler and Diffuser Fig.4.2(f), the recirculation zone is small, which has contributed to the even lesser drag out of all the cases. As seen from the images, it can be deduced that diffuser tends to decrease the recirculation zone to have a more teardrop shape. Thus, organizing the flow and eventually drag coefficient of the vehicle is reduced.

Figures 4.3 & 4.4 show the comparison of the drag coefficient and lift coefficient respectively for different cases.

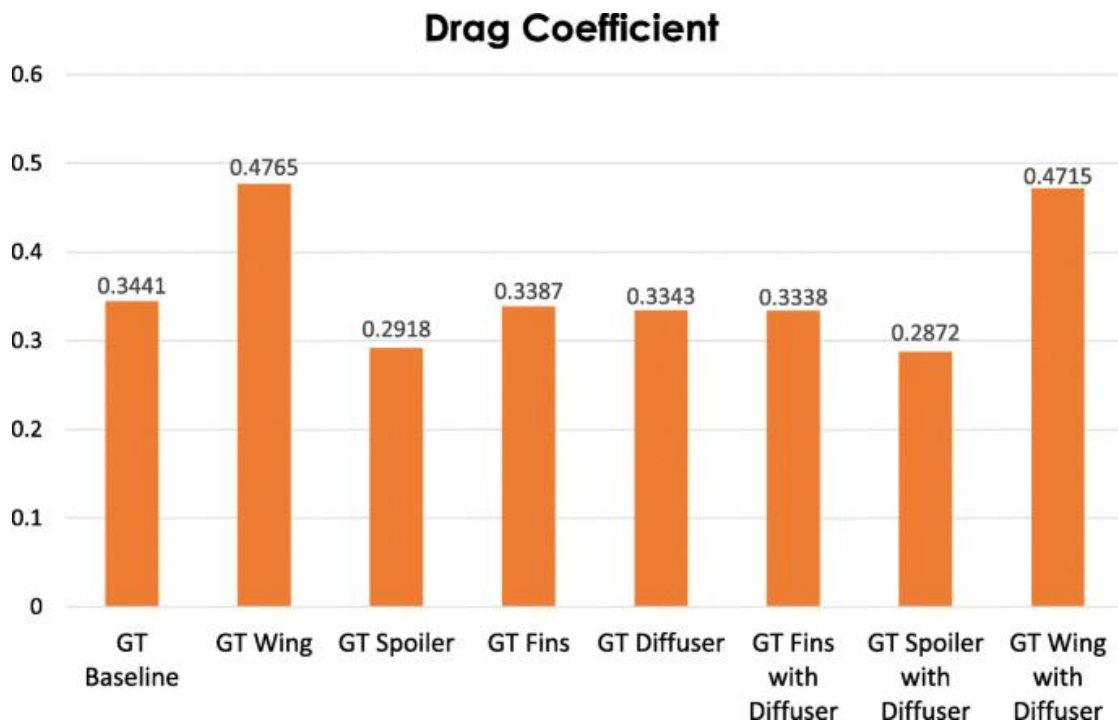


Figure 4.3 Comparison of the drag coefficient

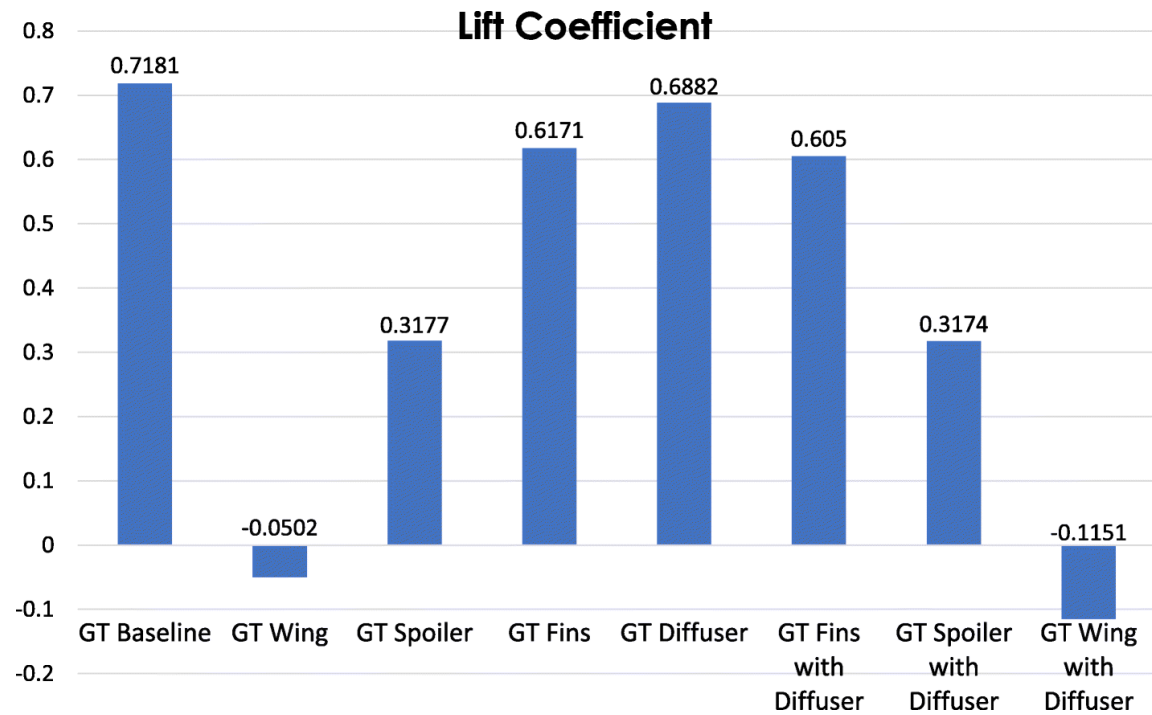


Figure 4.4 Comparison of the lift coefficient

From Table 4.c and Figs. 4.1 and 4.2, it can be found that the maximum drag is in the case of the GT Wing. By application of rear diffusers, the overall drag of GT Wing with diffuser is reduced by a little margin. Secondly, the drag in the Baseline GT was further reduced by diffuser by a considerable margin. The minimum drag is observed in the case of GT Spoiler with diffuser.

Table 4.a Drag coefficient, Lift coefficient and percentage reduction from the baseline GT model

S.No.	Model	Drag Coefficient	Lift Coefficient	Drag Coefficient Reduction (%)	Lift Coefficient Reduction (%)
1.	GT Baseline	0.3441	0.7181	–	–
2.	GT Wing	0.4765	–0.0502	+ 38.47	– 106.99
3.	GT Spoiler	0.2918	0.3177	–15.19	– 55.75
4.	GT Fins	0.3387	0.6171	–1.56	–14.06
5.	GT Diffuser	0.3343	0.6882	–2.84	–4.16
6.	GT Fins with Diffuser	0.3338	0.605	–2.99	–15.74
7.	GT Spoiler with Diffuser	0.2872	0.3174	–16.53	–55.8
8.	GT Wing with Diffuser	0.4715	–0.1151	+ 37.02	– 116.02

Table 4.b and Fig.4.5 shows the trend in drag force for different speeds. Three different speeds are considered 70 kmph, 150 kmph and 300 kmph for considering multiple scenarios. The graph validates the theory that drag force increases exponentially with the increase in speed.

Table 4.b Drag Force comparison at different speeds

S.No.	Model	Drag Force at 70 kmph (N)	Drag Force at 150 kmph (N)	Drag Force at 300 kmph (N)
1.	GT Baseline	159.22	731.23	2924.58
2.	GT Wing	220.48	1012.54	4049.88
3.	GT Spoiler	135.02	620.1	2480.07
4.	GT Fins	156.72	719.71	2878.68
5.	GT Diffuser	154.68	710.36	2841.29
6.	GT Fins with Diffuser	154.45	709.47	2837.04
7.	GT Spoiler with Diffuser	132.89	610.31	2440.97
8.	GT Wing with Diffuser	218.17	1002.009	4007.38

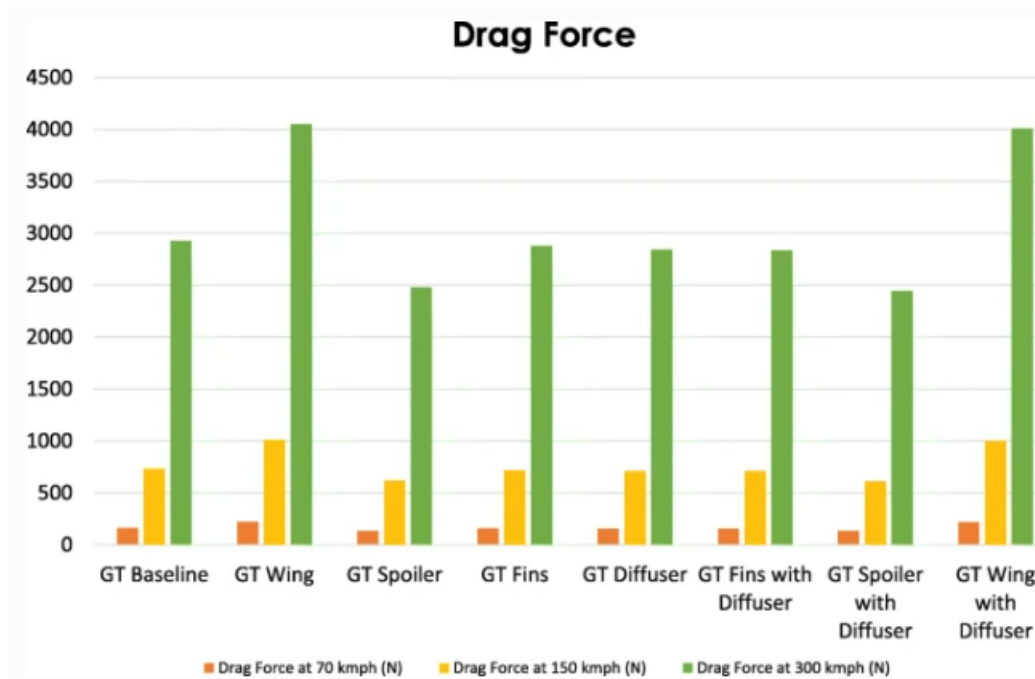


Figure 4.5 Comparison of the drag force at different speeds

Table 4.c and Fig.4.6 shows the trend in lift force for different speeds. Three different speeds are considered 70 kmph, 150 kmph and 300 kmph for considering multiple scenarios. The graph shows that the lift force increases exponentially with the increase in speed, but in the case of GT Wing and GT Wing with Diffuser, it is observed that negative lift or downforce is more prominent. It is useful for having increased traction with road, which subsequently leads to increased cornering speed. But this will increase the drag and hinder achieving the potential top speed.

Table 4.c Lift Force comparison at different speeds

S.No.	Model	Lift Force at 70 kmph (N)	Lift Force at 150 kmph (N)	Lift Force at 300 kmph (N)
1.	GT Baseline	332.27	1525.89	6103.29
2.	GT Wing	-23.22	-106.8	-426.66
3.	GT Spoiler	147.005	675.17	2700.2
4.	GT Fins	285.54	1311.31	5244.87
5.	GT Diffuser	318.44	1462.42	5849.16
6.	GT Fins with Diffuser	279.94	1285.53	5142.03
7.	GT Spoiler with Diffuser	146.86	674.49	2697.65
8.	GT Wing with Diffuser	-53.25	-244.73	-978.26

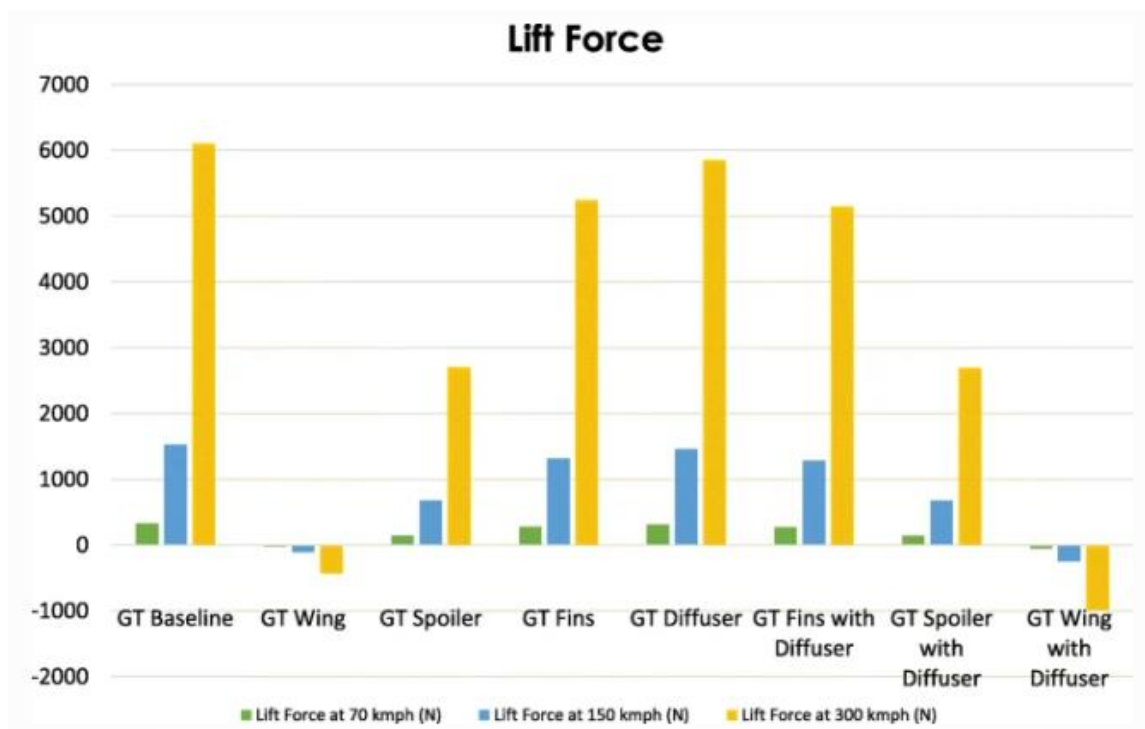


Figure 4.6 Comparison of the lift force at different speeds

Chapter 5

Future Scope

Aerodynamic simulation of vehicles and after adding accessories has a promising future, as advancements in computational power and simulation technology continue to improve the accuracy and efficiency of these simulations.

One area where this technology is likely to be applied is in the development of more efficient and sustainable automobiles. By simulating the aerodynamic performance of different body shapes and accessories, engineers can design cars that are more streamlined and have lower drag coefficients, reducing fuel consumption and emissions.

Another potential application is in the field of autonomous vehicles. As self-driving cars become more prevalent, their ability to navigate safely and efficiently will be influenced by their aerodynamic performance. By simulating different driving scenarios and environmental conditions, engineers can optimize the design of autonomous vehicles for maximum safety and efficiency.

Overall, the future scope of aerodynamic simulation for vehicles is vast, and the technology is likely to continue advancing rapidly in the coming years. As such, we can expect to see a wide range of applications for this technology across the automotive industry, from design and development to testing and optimization.

Chapter 6

Conclusion

The analysis of the stock GT with different aerodynamic accessories was simulated by using numerical simulation in this report. It was found that aerodynamic drag is influenced by using different accessories.

In consideration to reduce drag, it is beneficial that the flow is attached to the vehicle's body for as long as possible. A streamlined body would result in less flow separation, which would cause decrease in turbulence. In the case of GT with Spoiler and Diffuser, maximum of all drag reduction of 16.53% was observed. Although the fins indeed reduced drag to a certain extent, but they have a different functionality such as high speed stability by channeling airflow at the rear correspondingly.

Wing having an altogether different motive. It increased the drag by 27.78% but its priority function is to provide downforce in trade off to increased drag, and increase track performance. On the other hand, the diffusers decreased the drag when attached in all the different cases.

In conclusion, it may be regarded as proper optimization can lead to better aerodynamics of the vehicle in different scenarios.

Chapter 7

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